

ROBOT BASE – CHECKLIST

Team _____

Team Lead:

Team Members:

Submit this checklist + shared Fusion file + photos by _____

CAD FILE (Fusion 360)

- ☐ Deck component: parametric sketch, all features derived from parameters (e.g., two motor mount holes spaced 60mm, mirrored across centerline with 50mm offset)
- ☐ Every sketch is FULLY CONSTRAINED (all lines black, no blue)
- ☐ Custom components are “autonomous,” they don’t rely on sketches or bodies located in other parts
- ☐ 8 custom parts modeled & named properly:
 - ☐ Motor mount ×2 Assigned: _____
 - ☐ Battery holder Assigned: _____
 - ☐ Protoboard mount Assigned: _____
 - ☐ KCD1 switch holder Assigned: _____
 - ☐ Wago connector block Assigned: _____
 - ☐ IR-sensor mount Assigned: _____
 - ☐ Deck (polycarbonate) Assigned: _____

ASSEMBLY:

- ☐ Minimum 13 components total (8 custom parts + motors, wheels, etc.) inserted & fully constrained (0 degrees of freedom remaining: you can try to pull pieces apart but they don’t move)
- ☐ Wheels constrained to ground plane with 45mm offset
- ☐ Deck is production ready: CAM setup sheet

PHYSICAL BUILD

- ☐ Polycarbonate deck CNC-cut from (tolerance ± 0.25 mm on holes)
- ☐ ≥ 3 custom 3D-printed parts printed & fitted (photos required)
- ☐ Robot rolls straight ≥ 1 m on floor without wobble or binding
- ☐ Battery, protoboard, switch, Wago terminals, and IR sensor are mounted and wired
- ☐ 5S photo: station clean, tools returned, batteries charging

ITERATION LOG (Individual – already assigned)

- ☐ Each student has ≥ 3 dated entries with measurements (e.g., “v3 motor mount – interference 0.8 mm $\rightarrow$ added 1 mm parameter clearance”)
- ☐ At least one failed print photo + root-cause note

SUBMISSION FOLDER (Fusion 360 Group Project Hub)

- ☐ Folder: Photos (8+) + CNC .DXF + 3D-print .STL files + completed checklist PDF
- ☐ Fusion 360 project folder contains part iteration logs, CAM setup sheet, this checklist

